

the two outputs must have compatible dimensions and coordinate semantics. If the branches are independently reduced to $\mathbb{R}^{H'_A}$ and $\mathbb{R}^{H'_B}$ with $H'_A \neq H'_B$, the addition is not defined. Even if both branches retain the same number of components, independently selected channel identities may no longer correspond. Safe structured pruning therefore coordinates retained index sets across dependent paths or inserts an explicit projection that restores compatibility.

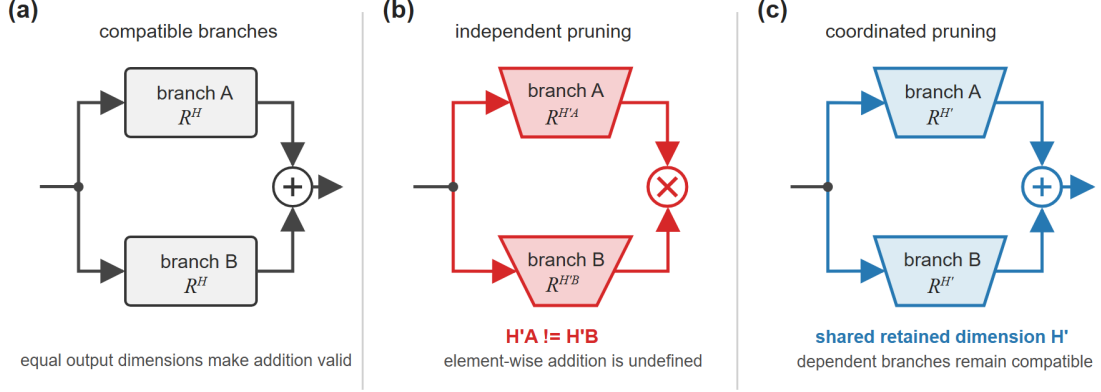


Figure 3.6: Dependency constraints in structured pruning. Panel (a) shows two compatible branches whose outputs can be added. Panel (b) illustrates an invalid result when the branches are independently reduced to different output dimensions. Panel (c) uses a shared retained structure so that both branches remain compatible. Similar consistency constraints apply between successive layers and between the coupled gates of recurrent networks.

Figure 3.6 illustrates why a structured pruning decision is not solely a local ranking problem. The removable unit is defined by the computation graph. For recurrent architectures, one hidden state participates in several gate matrices, the recurrent state vector, and the input of a subsequent output head. Treating these occurrences as one group preserves a valid recurrent model after removal. This architectural coupling is revisited for the LSTM estimator in Section 7.4.

Pruning criteria and ranking scope

After the pruning structure has been defined, an importance score is required for every candidate group. The score estimates the expected effect of removing the candidate. No inexpensive score can prove global optimality because parameters interact and the loss changes after each removal. Criteria therefore trade computational cost against task-specific information.